BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Paper I (Objective Type) — Ist - A - Exam - 2023

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6195

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. If A = [-2 1; 3 5] then A - A^t is :

(A) [0 2; -2 0]
(B) [0 4; 4 10]
(C) [0 -2; 2 0]
(D) [-2 0; 0 2]

Q2. Venn Diagrams are useful only in case of :

(A) Concrete Sets
(B) Abstract Sets
(C) Subsets
(D) Universal Sets

Q3. If Z = 2 - 3i, then |Z|^2 =

(A) sqrt(5)
(B) 5
(C) sqrt(13)
(D) 13

Q4. If A is a Square Matrix of Order 3, then |KA| = :

(A) 3|A|
(B) 9|A|
(C) K|A|
(D) K^3|A|

Q5. An equation in which two Algebraic Expressions are equal for particular values of the variable is called :

(A) An Equation
(B) Conditional Equation
(C) Identity
(D) Both A and B

Q6. If for a Quadratic Equation ax^2 + bx + c = 0, b^2 - 4ac = 0, then roots of the equation will be :

(A) Rational
(B) Irrational
(C) Equal
(D) Unequal

Q7. The Sum of the Roots of the Equation 5x^2 - x - 2 = 0 is :

(A) -2/5
(B) 2/5
(C) -1/5
(D) 1/5

Q8. If an = (-1)^n(2n - 3), then 5th term of the sequence is :

(A) -7
(B) 7
(C) 49
(D) -49

Q9. A Coin is tossed four times, then the Probability of getting no head is :

(A) 1/16
(B) 1/8
(C) 1/4
(D) 1/2

Q10. The Number of 4 Digit Numbers that can be formed out of digits 1, 2, 3, 4, 5, 6 when no digit is repeated is :

(A) 15
(B) 36
(C) 360
(D) 720

Q11. H.M. between 1/a and 1/b is :

(A) 2ab/(a+b)
(B) (a+b)/2ab
(C) 2/(a+b)
(D) (a+b)/2

Q12. 1 + 3 + 5 + ------ + (2n + 5) = (n + 3)^2 is true for :

(A) n >= -1
(B) n >= -2
(C) n >= 1
(D) n >= 2

Q13. Cos(2x + 30 deg).Cos(2x - 30 deg) = :

(A) 1/2[Sin4x - Sin60 deg]
(B) -1/2[Sin4x + Sin60 deg]
(C) 1/2[Cos4x + Cos60 deg]
(D) 1/2[Cos4x - Cos60 deg]

Q14. Which of the following is a pair of Coterminal Angles :

(A) 30 deg, -330 deg
(B) 50 deg, -330 deg
(C) 30 deg, 760 deg
(D) 60 deg, 1480 deg

Q15. The Coefficients of the terms equidistant from beginning and end of the expansion of (a + x)^n; n in N are equal as :

(A) (n r) = (n n-r)
(B) (n r) = (n n+r)
(C) (n r+1) = (n r)
(D) (n r) = (n-1 r-1)

Q16. Range of y = 3Sin2x is :

(A) [-1, 1]
(B) [-3, 3]
(C) [-5, 5]
(D) [-6, 6]

Q17. Sin^-1(1/sqrt(2)) = ------ :

(A) 0
(B) pi/6
(C) pi/4
(D) pi/3

Q18. abc/R = :

(A) 4rs
(B) 4 Delta
(C) 4/Delta r
(D) Delta r/4

Q19. Area of a Triangle ABC is equal to :

(A) 1/2 bc Sinalpha
(B) 1/2 ab Sinalpha
(C) 1/2 bc Cosalpha
(D) 1/2 ac Singamma

Q20. Reference Angle of Cosx = -1/2 is :

(A) pi/6
(B) pi/4
(C) pi/2
(D) pi/3

Mathematics (Subjective) — Ist - A - Exam - 2023

Time Allowed: 2:50 Hours | **Maximum Marks:** 80

Note: Note: It is compulsory to attempt any (8 - 8) Parts each from Q.No.2 and Q.No.3 while attempt any (9) Parts from Q.No.4. Attempt any (3) Questions from Part - II. Write same Question No. and its Part No. as given in the Question Paper — Part - I — 25 x 2 = 50

Q2.

Q.No.2

(i) Prove that: a/b = c/d iff ad = bc

(ii) Show that for all Z in C, Z^2 + conjugate(Z)^2 is a real number.

(iii) Show A - B and B - A by Venn Diagram, when A and B are Overlapping Sets.

(iv) Verify the Commutative Property of Union and Intersection for the sets A = {1, 2, 3, 4, 5}, B = {4, 6, 8, 10}

(v) Construct Truth Table for the Statement (p -> ~p) v (p -> q)

(vi) If A = [2 -1; 3 1] verify that (A^-1)^t = (A^t)^-1

(vii) Find value of 'x' if: |1 2 1; 2 x 2; 3 6 x| = 0

(viii) Solve the Matrix Equation 2X - 3A = B if A = [1 -1 2; -2 4 5] and B = [3 -1 0; 4 2 1]

(ix) Reduce to Quadratic Form (x + 1)(x + 2)(x + 3)(x + 4) = 24

(x) Show that: x^3 - y^3 = (x - y)(x - wy)(x - w^2 y)

(xi) Show that the roots of the equation (b - c)x^2 + (c - a)x + (a - b) = 0; a, b, c in Q will be real.

(xii) If the roots of the equation x^2 - px + q = 0 differ by unity, prove that p^2 = 4q + 1

Q3.

Q.No.3

(i) Define Proper Rational Fraction.

(ii) Write the first four terms of the sequence if an - an-1 = n + 2, a1 = 2

(iii) If 5, 8 are two A.Ms between 'a' and 'b', find 'a' and 'b'

(iv) Sum the Series -8 - 3(1/2) + 1 + ------ + a11

(v) Find the 11th term of the Sequence 1 + i, 2, 2(1 - i), ------

(vi) Insert three G.Ms. between 2 and 32

(vii) Find the number of the Diagonals of a 6-sided figure.

(viii) Show that: 16C11 + 16C10 = 17C11

(ix) Two Dice are thrown. What is the Probability that the sum of the number of dots appearing on them is 4 or 6?

(x) If Sn = n(2n - 1), then find the Series.

(xi) Expand upto four terms (1 + x)^(-1/3)

(xii) If 'x' is so small that its square and higher powers can be neglected then show that (1 - x)/sqrt(1 + x) approx 1 - 3/2 x

Q4.

Q.No.4

(i) A Railway Train is running on a Circular Track of Radius 500 meters at the rate of 30 Km per hour. Through what angle will it turn in 10 Sec?

(ii) Verify Cos2theta = Cos^2 theta - Sin^2 theta when theta = 30 deg, 45 deg

(iii) Find the period of Sec 9x.

(iv) Prove that: sqrt((1 - Sinttheta)/(1 + Sintheta)) = Sectheta - Tantheta, where theta is not an odd multiple of pi/2

(v) Prove that: Sin780 deg Sin480 deg + Cos120 deg Sin30 deg = 1/2

(vi) Prove that: (Cos11 deg + Sin11 deg)/(Cos11 deg - Sin11 deg) = Tan56 deg

(vii) Prove that: Sin2alpha/(1 + Cos2alpha) = tanalpha

(viii) Find the Area of the Triangle ABC for given two sides and their included angle a = 200, b = 120, gamma = 150 deg

(ix) Show that r1 = S tan(alpha/2)

(x) Prove that: tan(gamma/2) = sqrt((s - a)(s - b)/s(s - c))

(xi) Without using calculator show that: 2Cos^-1(4/5) = Sin^-1(24/25)

(xii) Solve Cosec^2 theta = 4/3, theta in [0, 2pi]

(xiii) Find the value of theta, 2Sin^2 theta - Sin theta = 0, theta in [0, 2pi]

> Note: Part - II — 3 x 10 = 30

Q5. (Marks 8) (a) Show that |1 1 1; a b c; a^2 b^2 c^2| = (a - b)(b - c)(c - a) (5) (b) Solve the system of Equations x + y = 5, x^2 + 2y^2 = 17 (5)

Q6. (Marks 8) (a) Resolve (3x - 11)/((x^2 + 1)(x + 3)) into Partial Fractions (5) (b) A fair dice is thrown twice. Find the Probability that Prime Nos. of Dots appear in first throw and the Nos. of Dots in Second Throw is less than 5. (5)

Q7. (Marks 8) (a) The sum of the 6th and 8th terms of an A.P. is 40 and product of 4th and 7th term is 220, find the A.P. (5) (b) Use Binomial Theorem to show that: 1 + 1/4 + (1.3)/(4.8) + (1.3.5)/(4.8.12) + -------- = sqrt(2) (5)

Q8. (Marks 8) (a) Prove the Identity: 1/(Cosectheta - Cottheta) - 1/Sintheta = 1/ Sintheta - 1/(Cosectheta + Cottheta) (5) (b) Prove the Identity: (Cos3theta/Costheta) + (Sin3theta/Sintheta) = 4 Cos2theta (5)

Q9. (Marks 8) (a) Prove that: Delta = 4Rr Cos(alpha/2) Cos(beta/2) Cos(gamma/2) (5) (b) Prove that: Sin^-1(3/5) + Sin^-1(8/17) = Sin^-1(77/85) (5)